

Positive Feedback Loops as a Pathway to Planetesimal Formation

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PFITS+

Planet Formation in the Southwest +

3rd PFITS+ Meeting

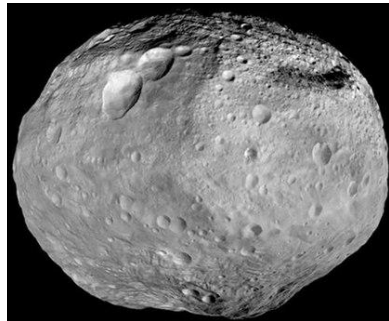
on Hydrodynamics of Circumstellar Disks and Planet Formation

Overview of planet formation

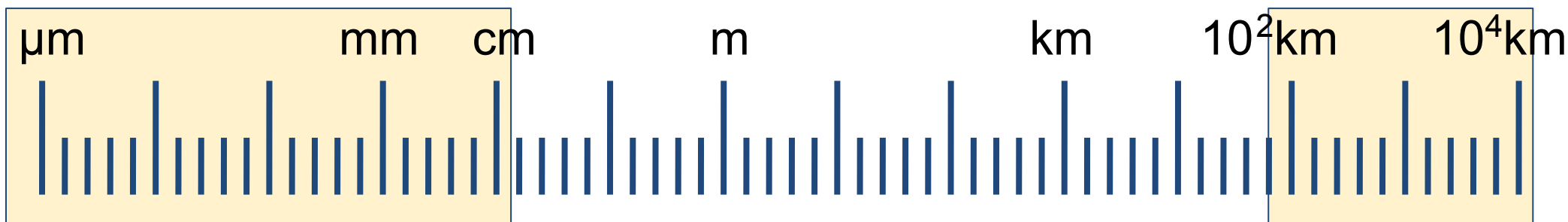
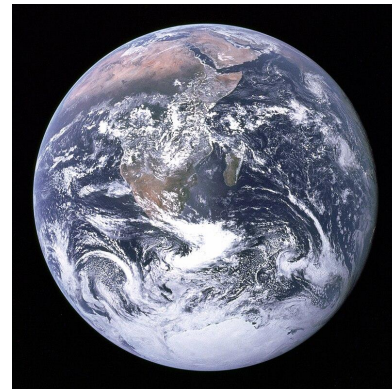
pebbles



planetesimals



planets



dust
coagulation

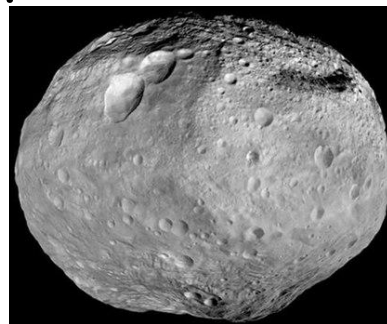
gravity

Overview of planet formation

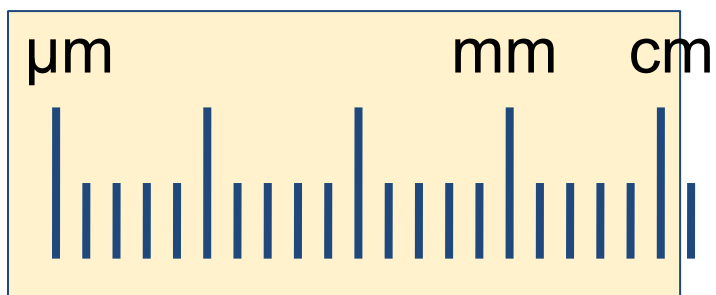
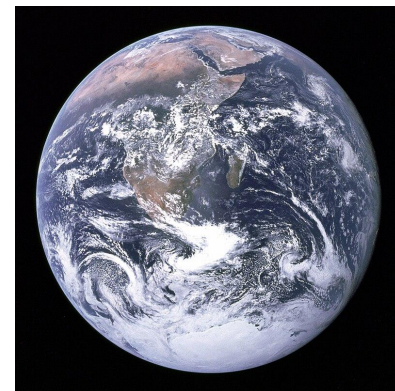
pebbles



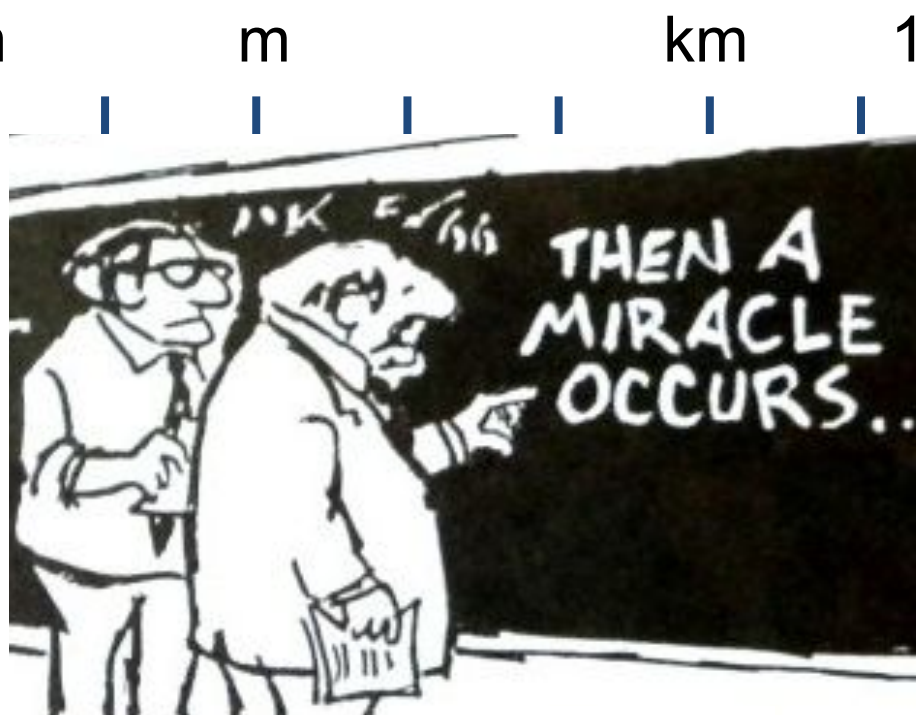
planetesimals



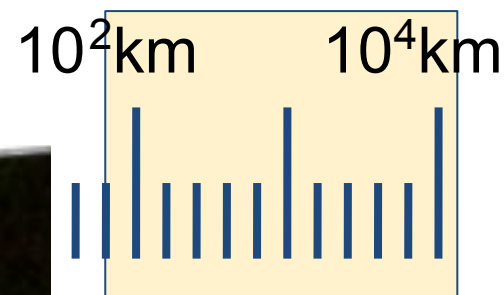
planets



dust
coagulation



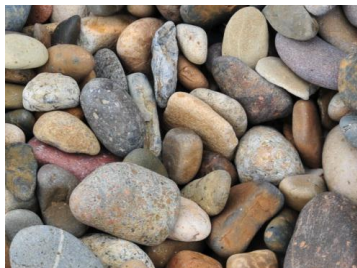
"I THINK YOU SHOULD BE MORE
EXPLICIT HERE IN STEP TWO."



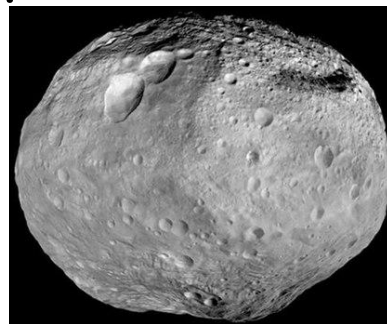
gravity

Overview of planet formation

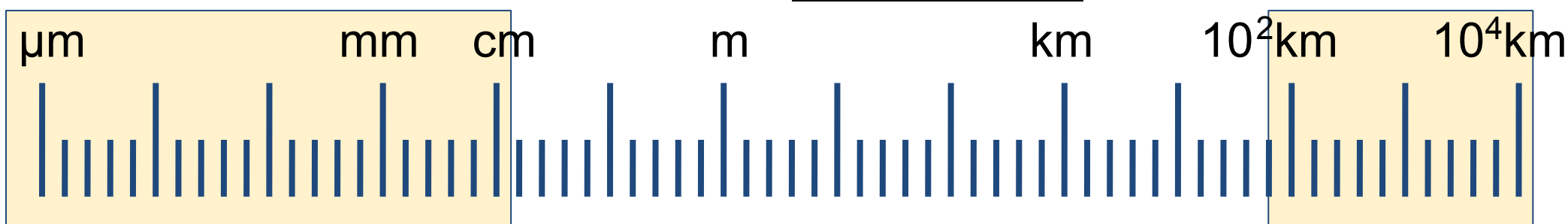
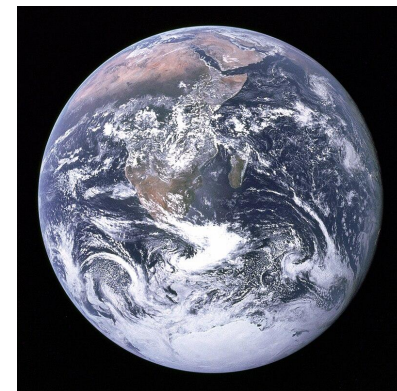
pebbles



planetesimals



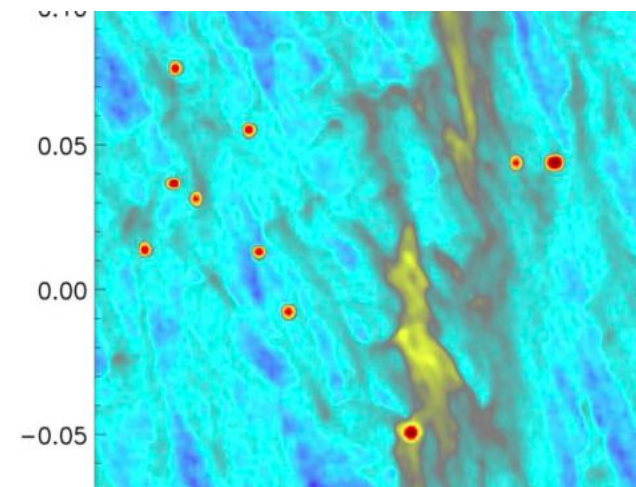
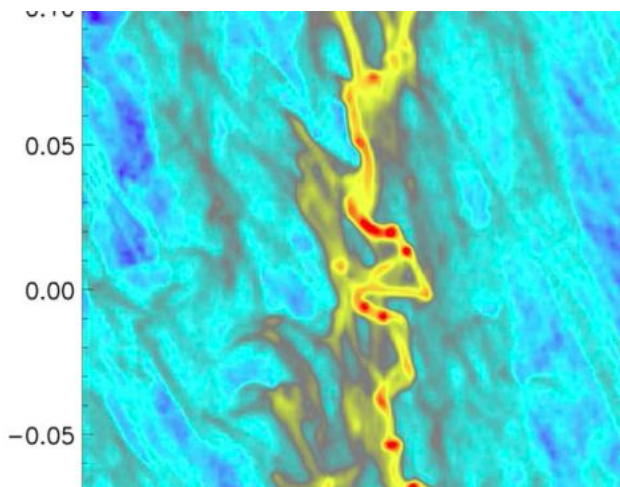
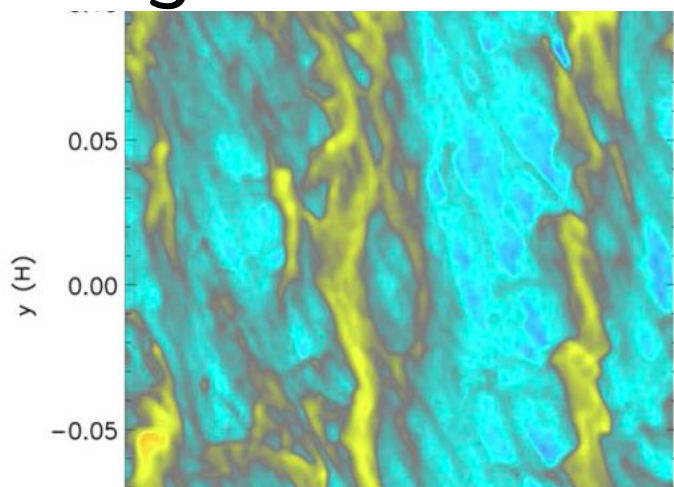
planets



dust
coagulation

Streaming Instability?
Vortices? Something else?

gravity



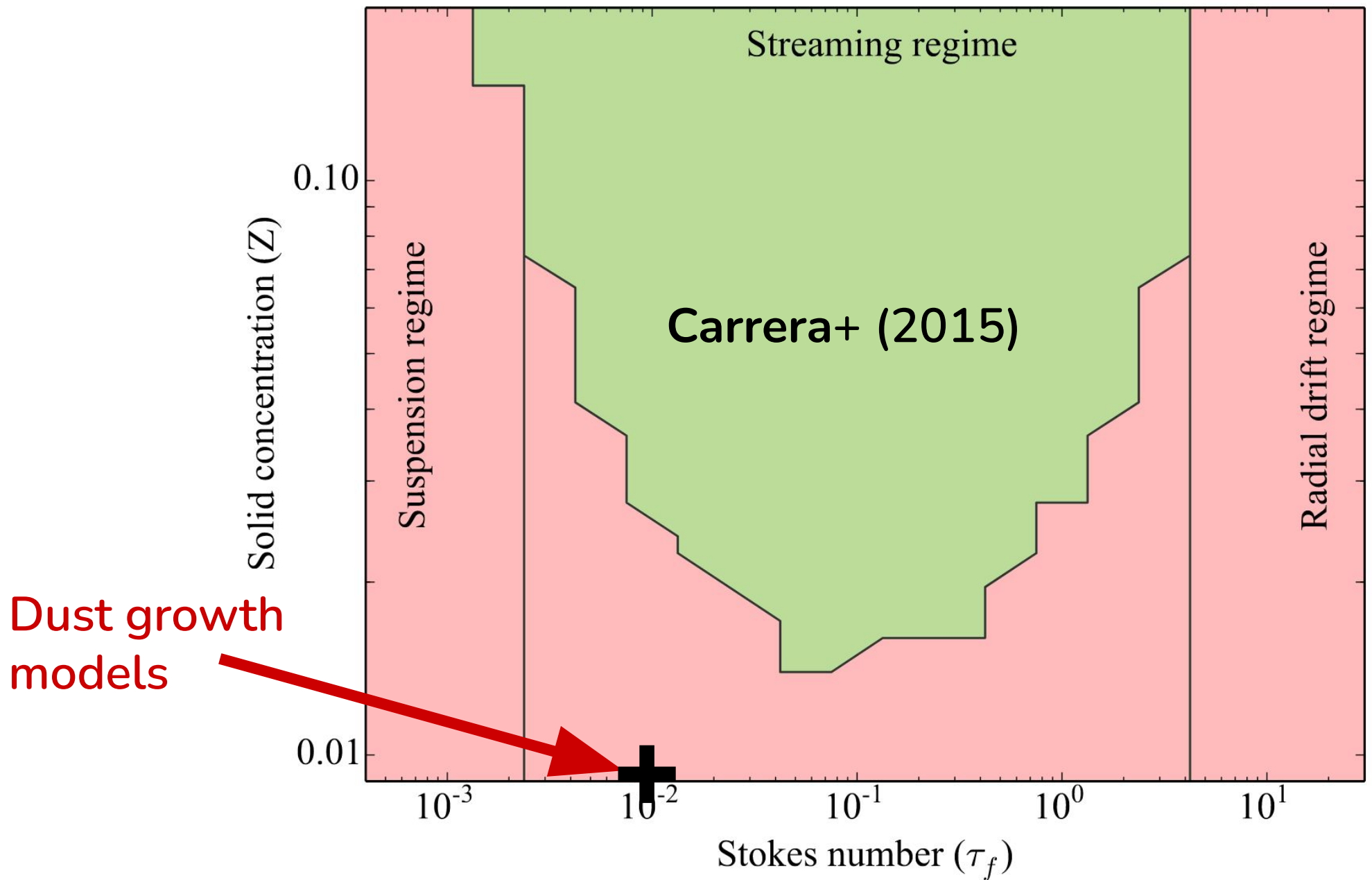
How to make **planetesimals** :

My Claims:

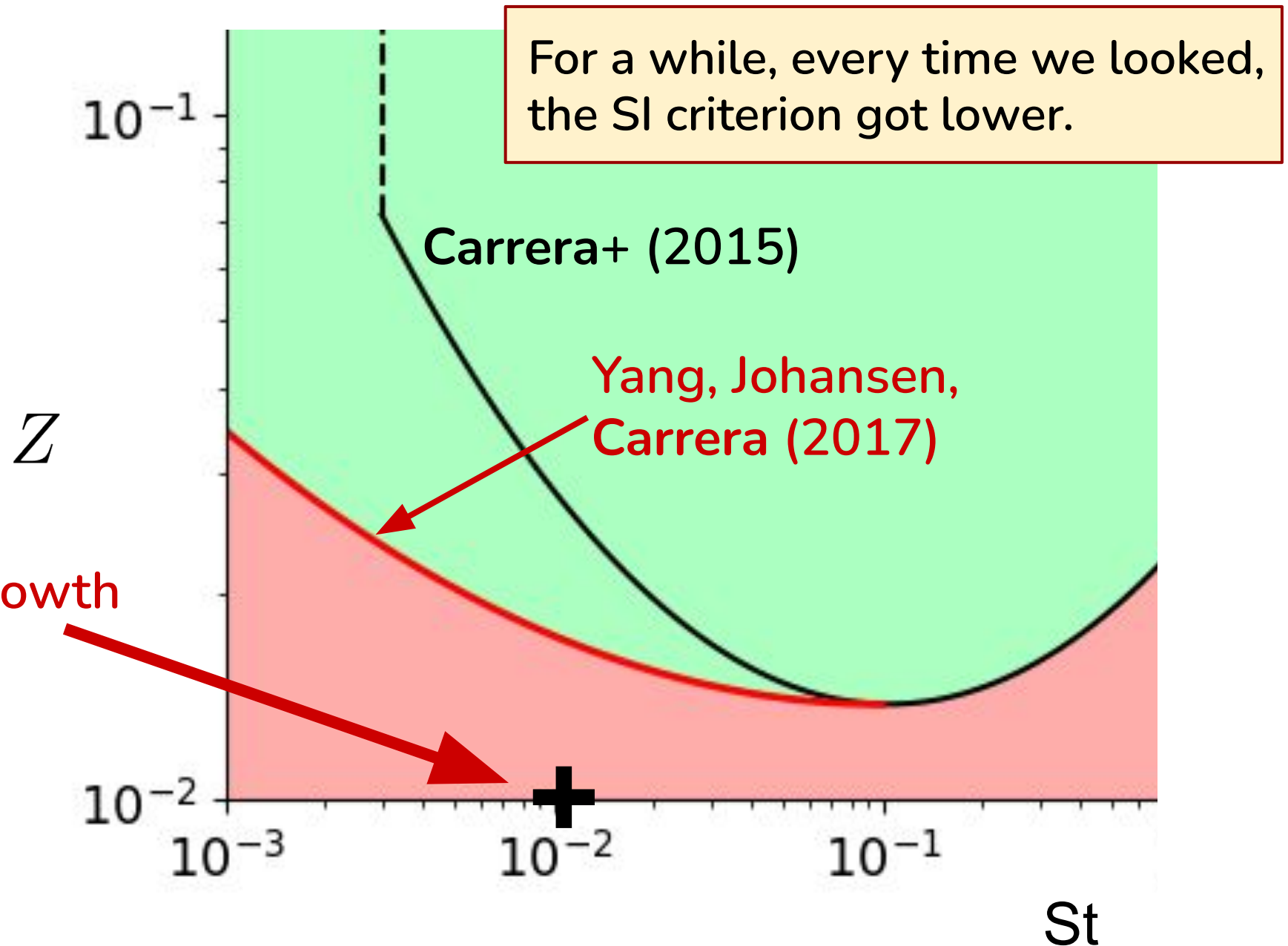
(1) We need to focus on small grains
(**$St \leq 0.01$**)

(2) We need to model the
Coagulation-Concentration Feedback Loop

History: SI Clumping Criterion

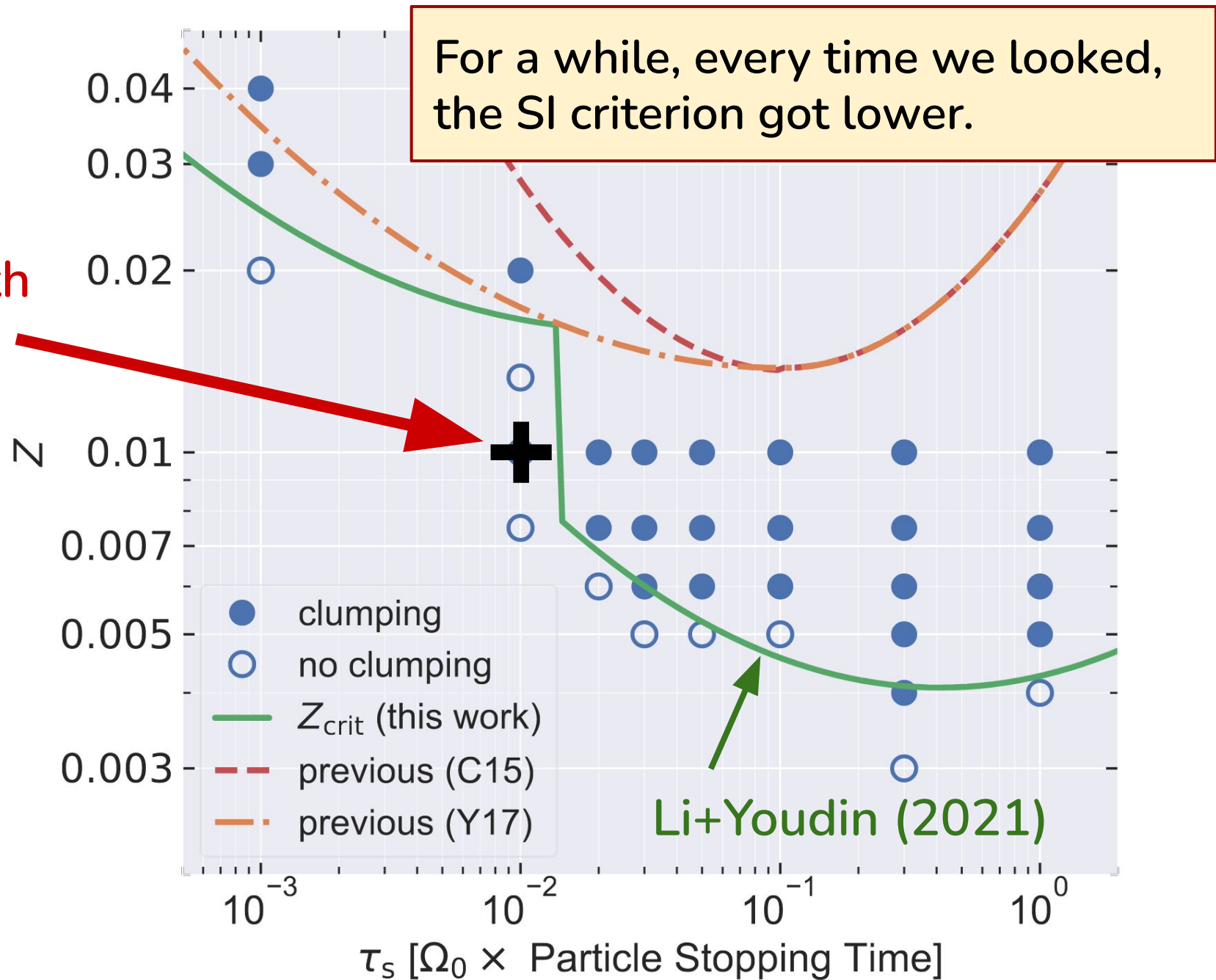


History: SI Clumping Criterion



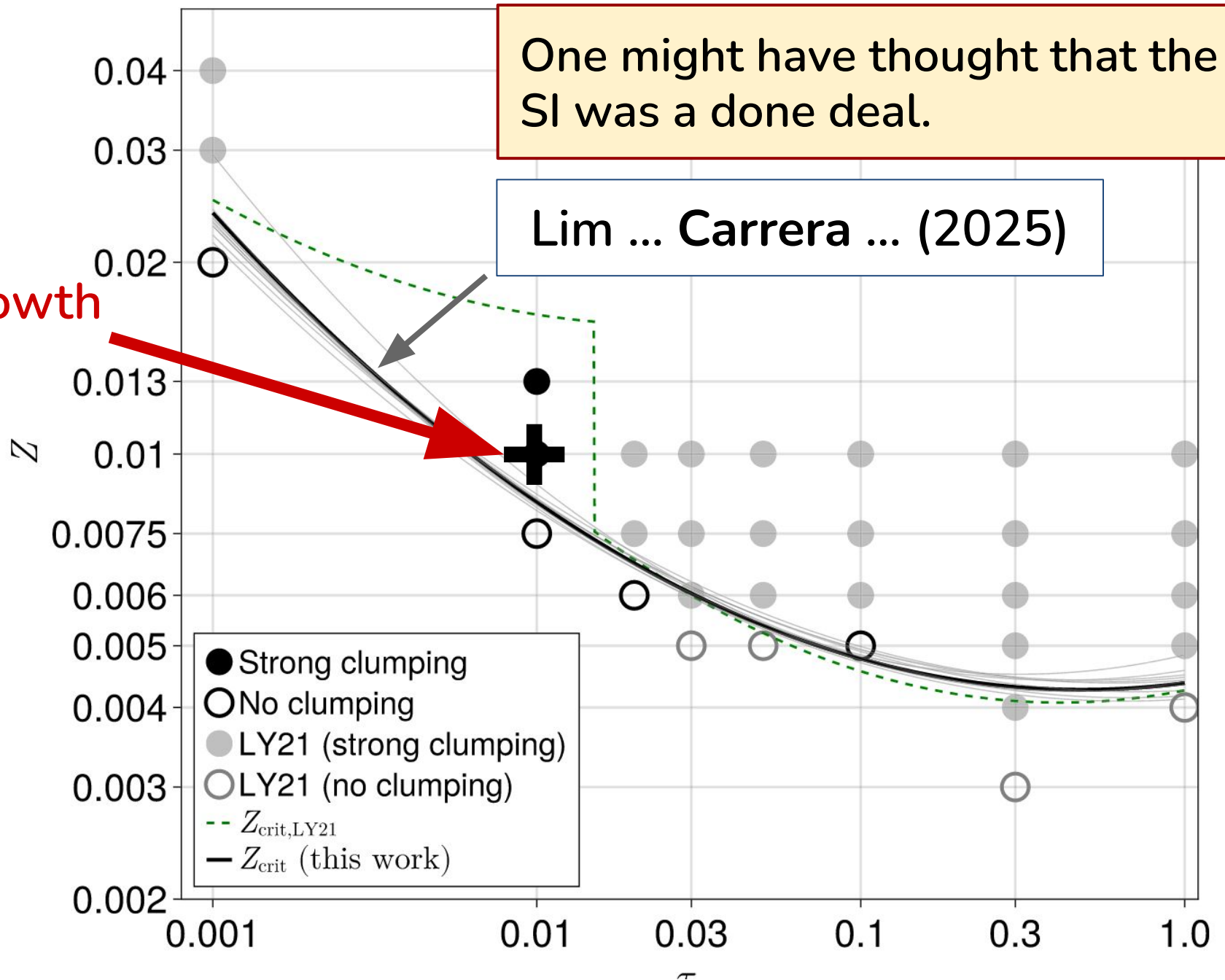
History: SI Clumping Criterion

Dust growth models

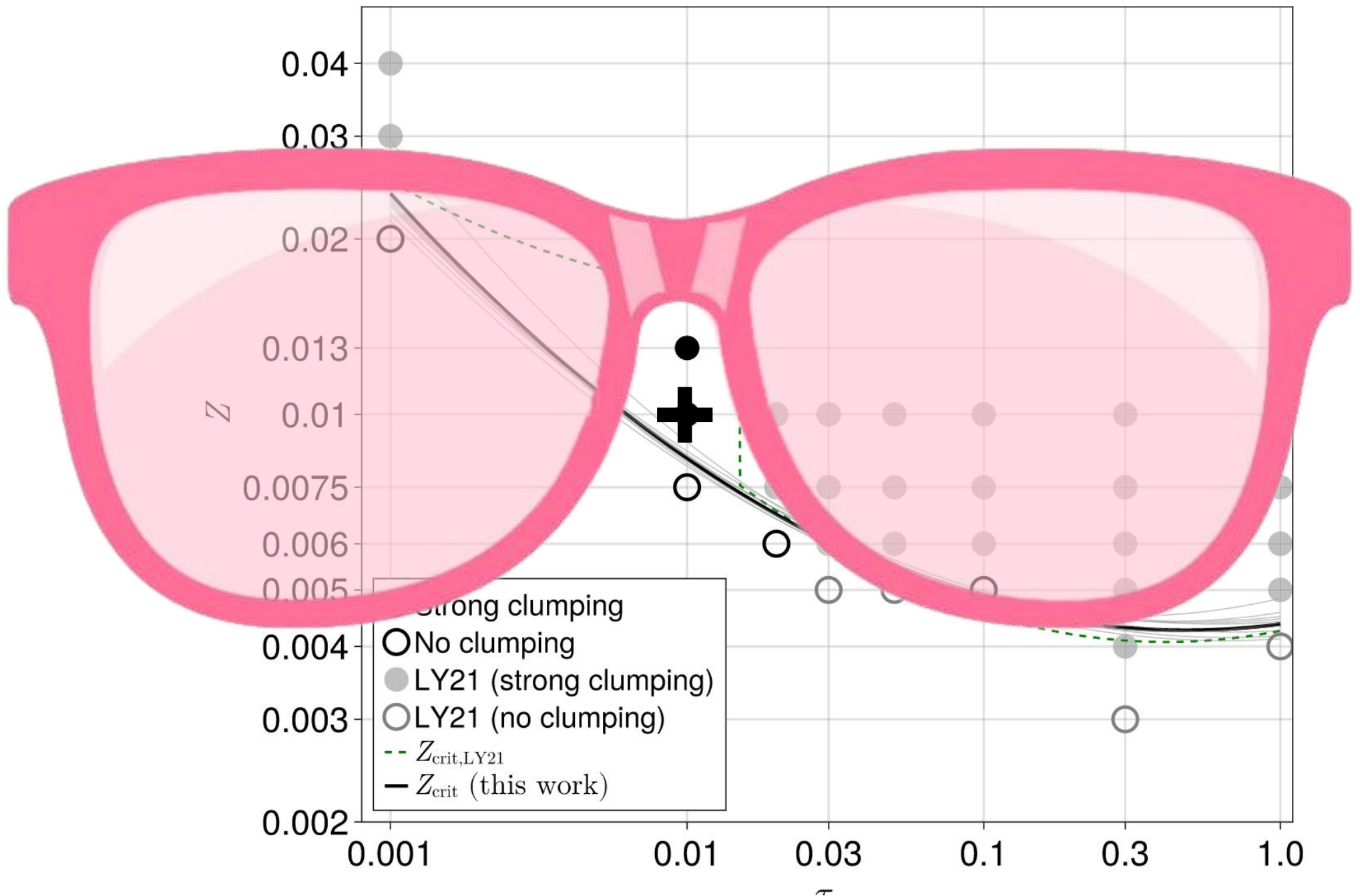


History: SI Clumping Criterion

Dust growth
models



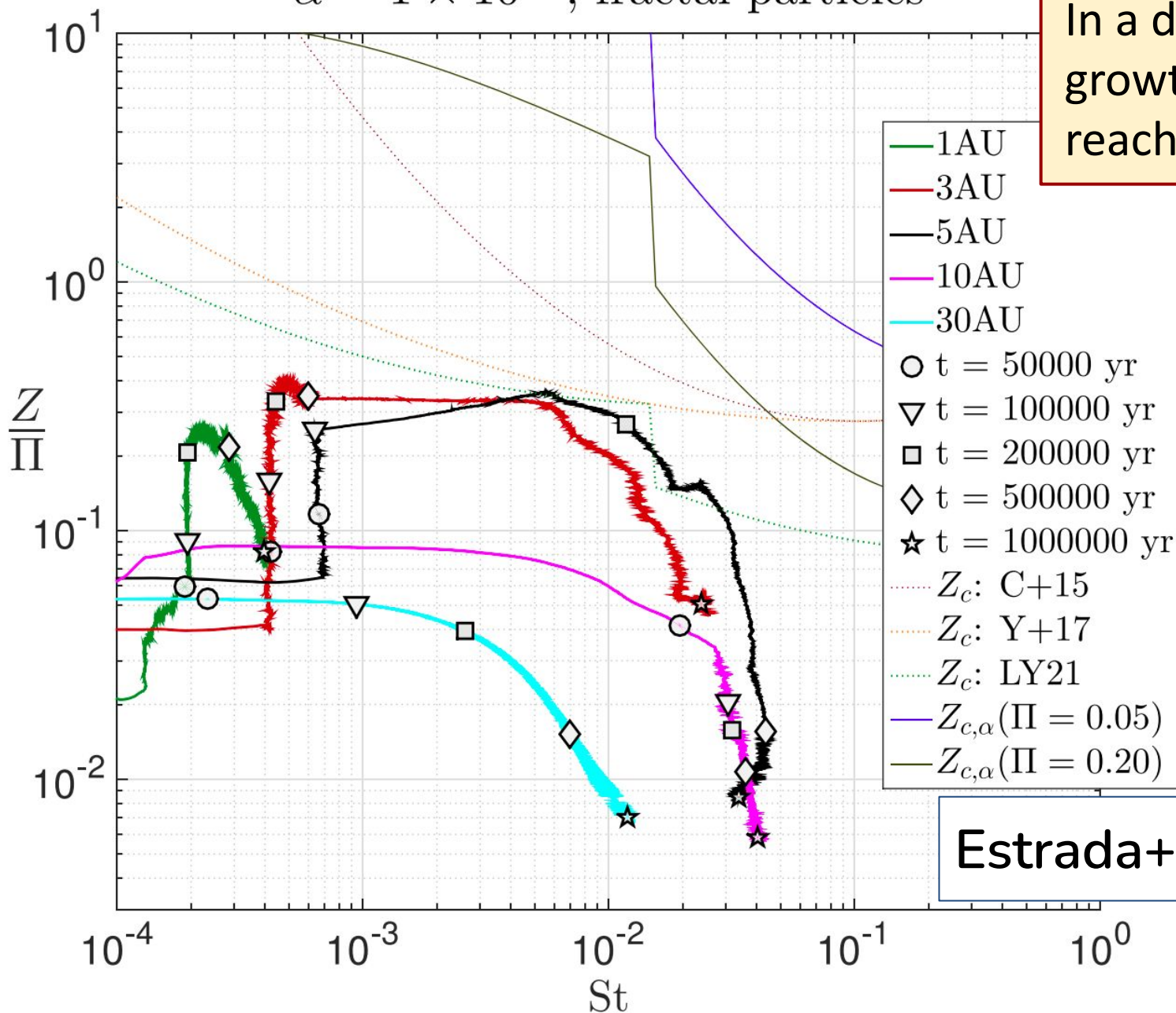
History: SI Clumping Criterion



Problem 1: Small Dust Grains

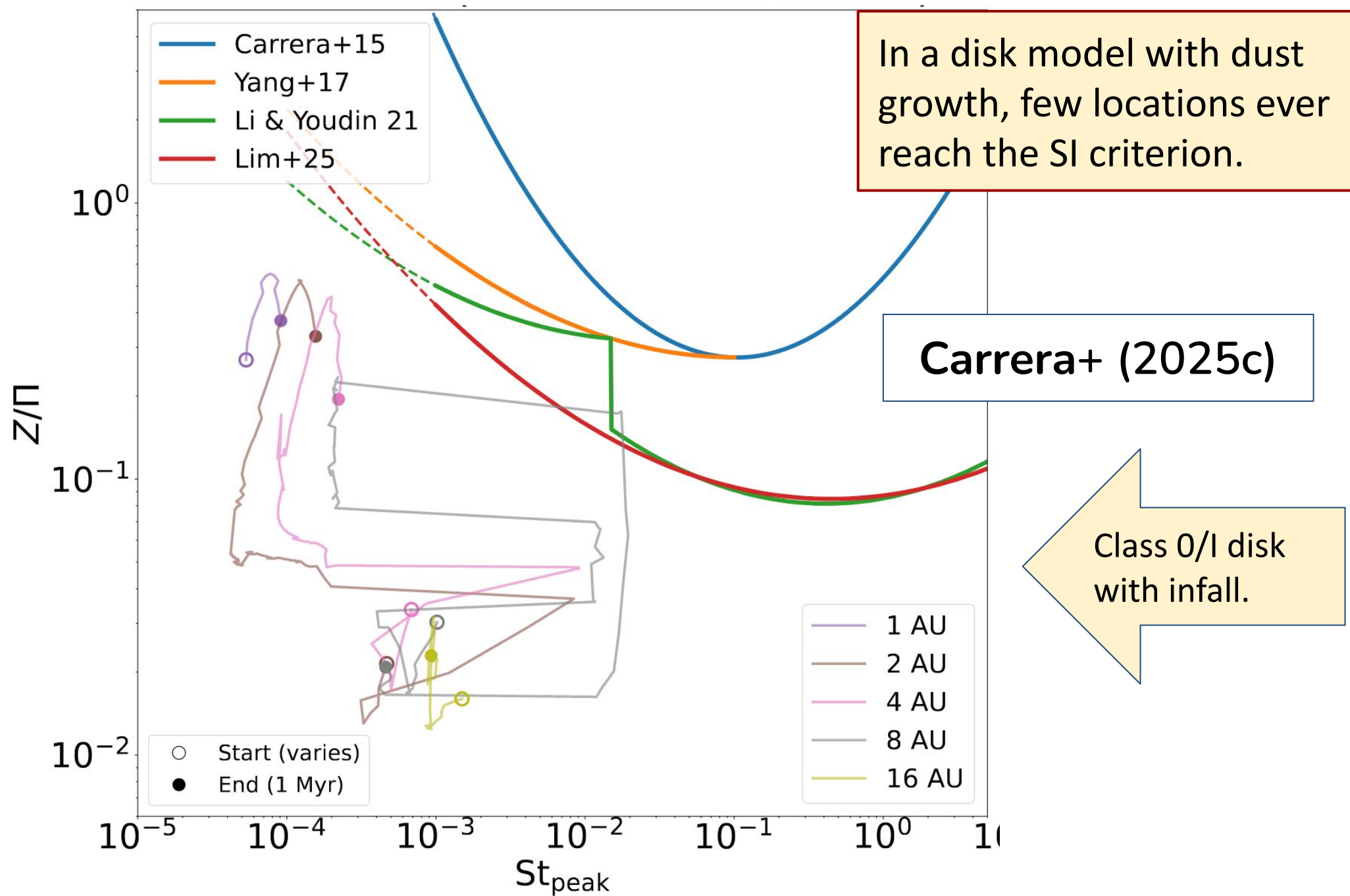
$\alpha = 1 \times 10^{-3}$, fractal particles

In a disk model with dust growth, few locations ever reach the SI criterion.



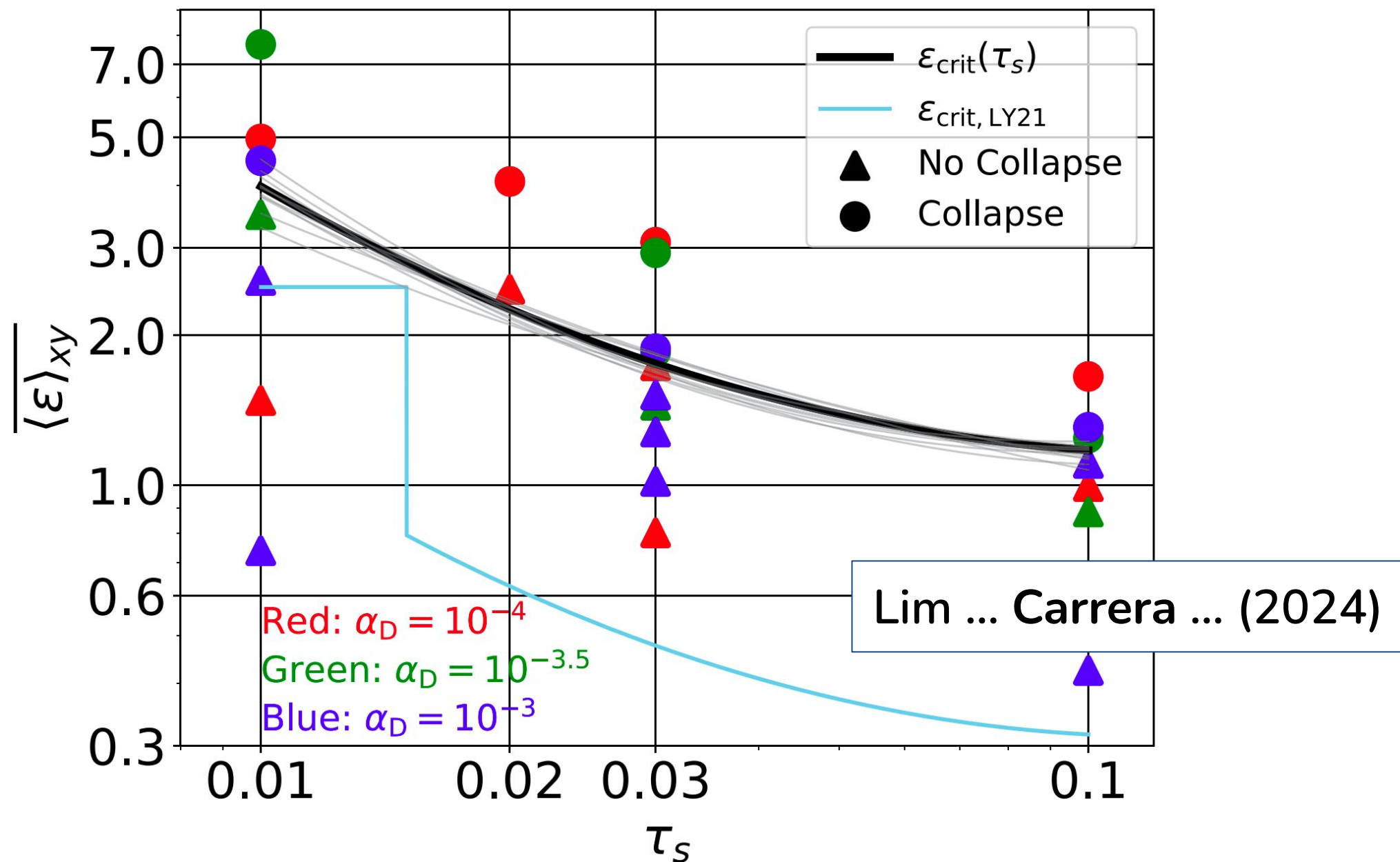
Estrada+Umurhan (2023)

Problem 1: Small Dust Grains



Problem 2: Turbulence

But it gets worse...
turbulence severely inhibits the SI.

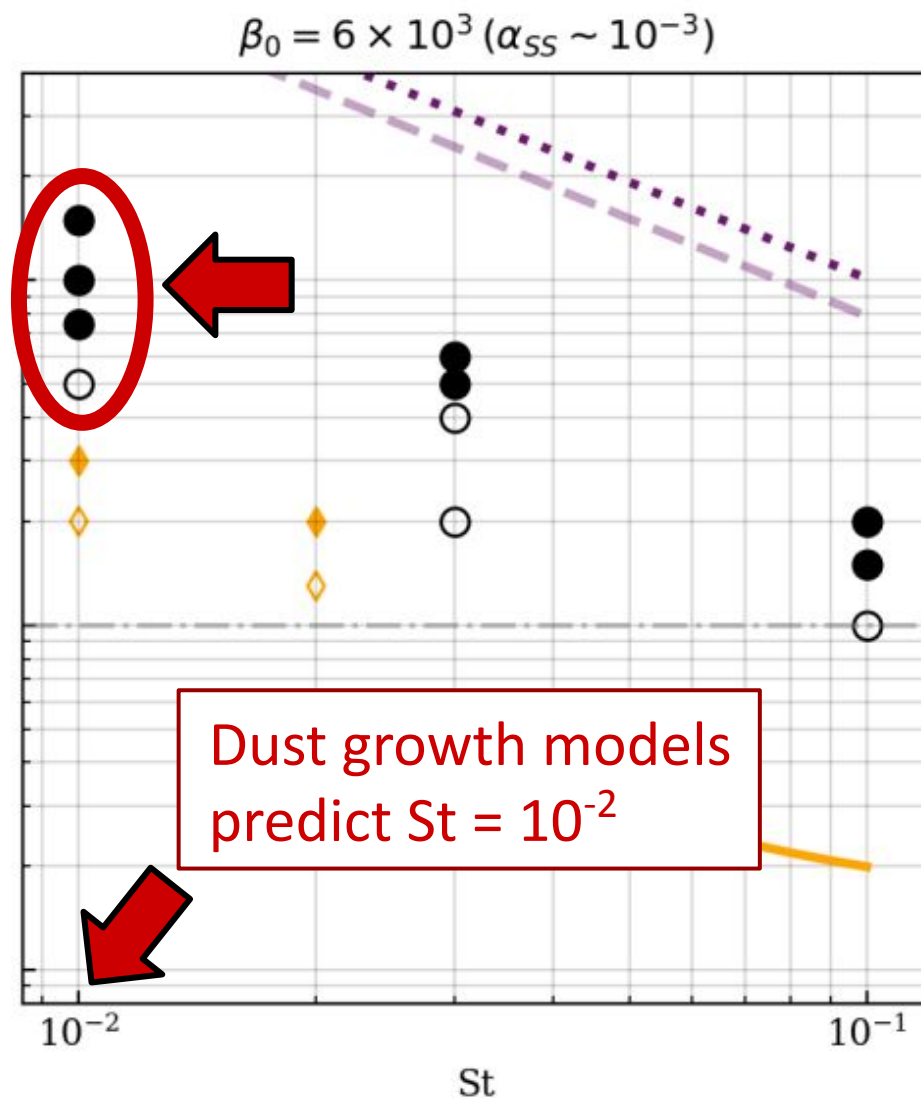
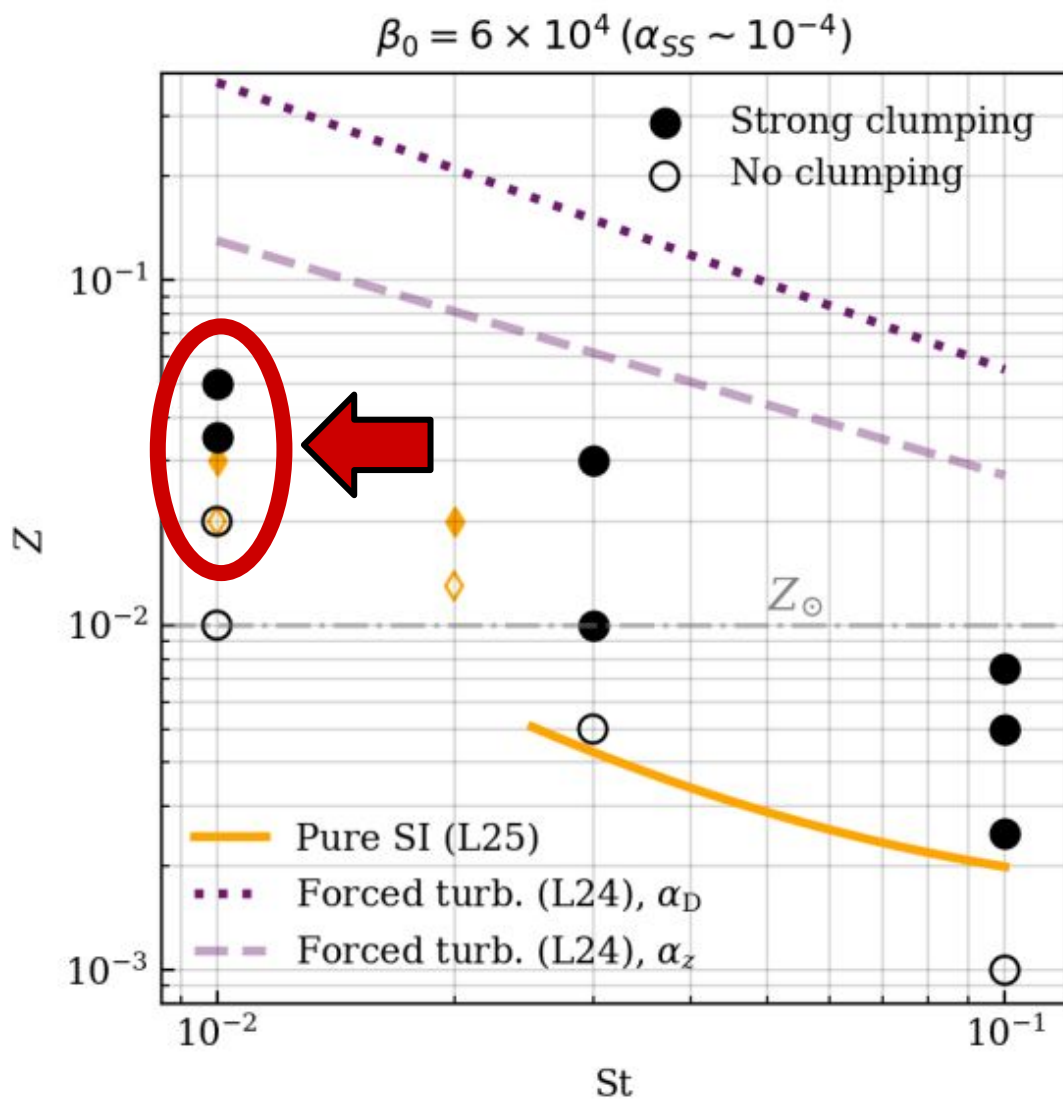


Problem 2: Turbulence

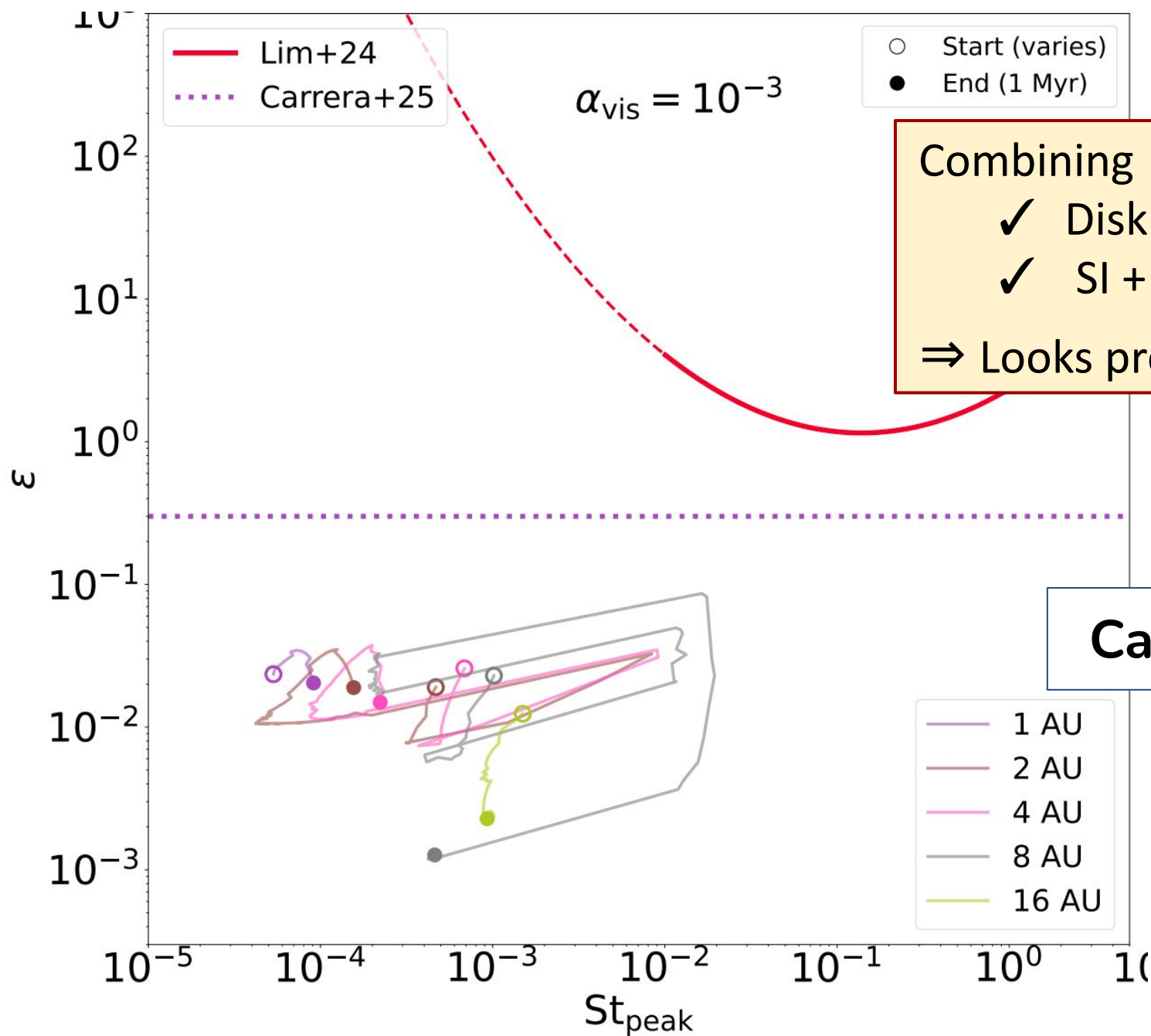
See also: Lin's talk (Monday)

Eriksson+ (2026)

The problem is less severe with MRI turbulence, up to a point.



Problem 2: Turbulence



Combining

- ✓ Disk model -- Class 0/I
- ✓ SI + turbulence criterion

⇒ Looks pretty hopeless

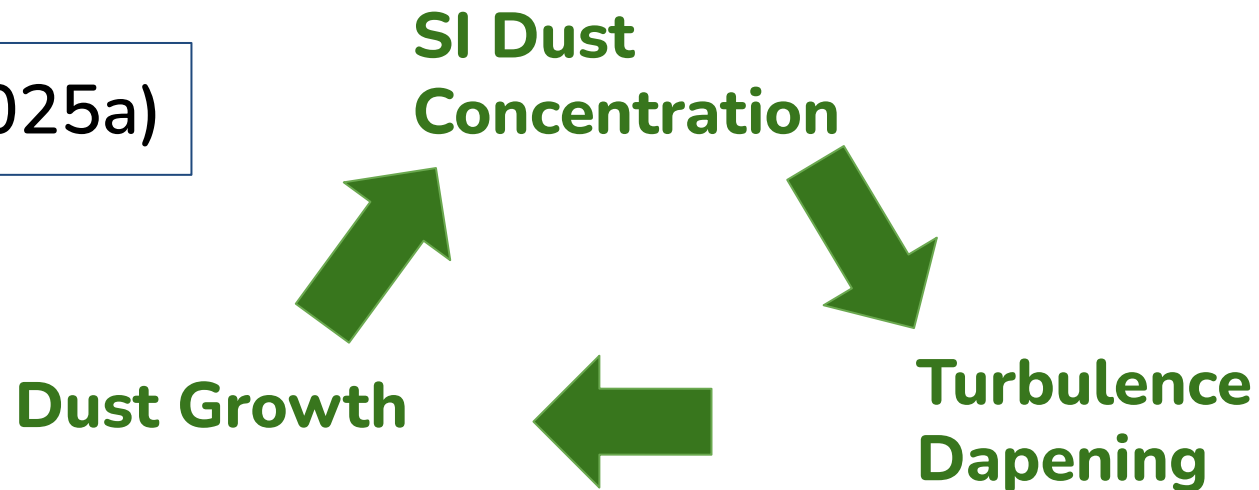
Carrera+ (2025c)

See also:

Orkan's talk
Debanjan's talk

Solution 1: Positive Feedback: SI+Coagulation

Carrera+ (2025a)



See also:

Ryosuke's talk (tomorrow)

Key insight:

The **streaming instability** and **dust growth** both alter the environment in a way that benefits the other.

Model Ingredients:

1. SI concentrates dust (from simulations)
2. Feedback dampens turbulence (analytic)
3. Lower turbulence = less fragmentation (analytic)

Solution 1: Positive Feedback: SI+Coagulation

Step 1: Dust concentration dampens turbulence

Most analytic models treat the gas+dust as a **colloid**.

(i.e. single fluid where dust adds to the mass but not to the pressure)

That is valid for $St \approx 0$ but not for larger St .

We used an **energy equipartition** argument to derive:

$$St_{\text{frag}} = \frac{St_x - 1 + \sqrt{(St_x + 1)^2 + 4\epsilon St_x}}{2}, \quad \text{where } St_x \equiv \frac{v_{\text{frag}}^2}{3\alpha c_s^2}.$$

Carrera+ (2025a)

Solution 1: Positive Feedback: SI+Coagulation

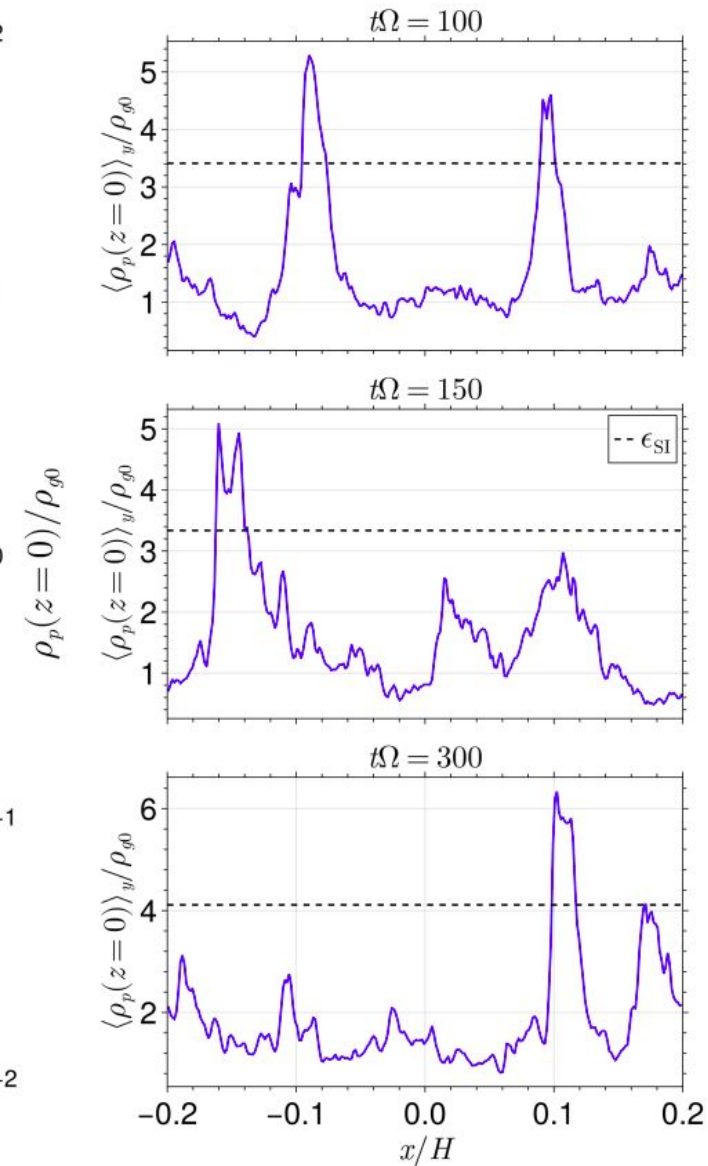
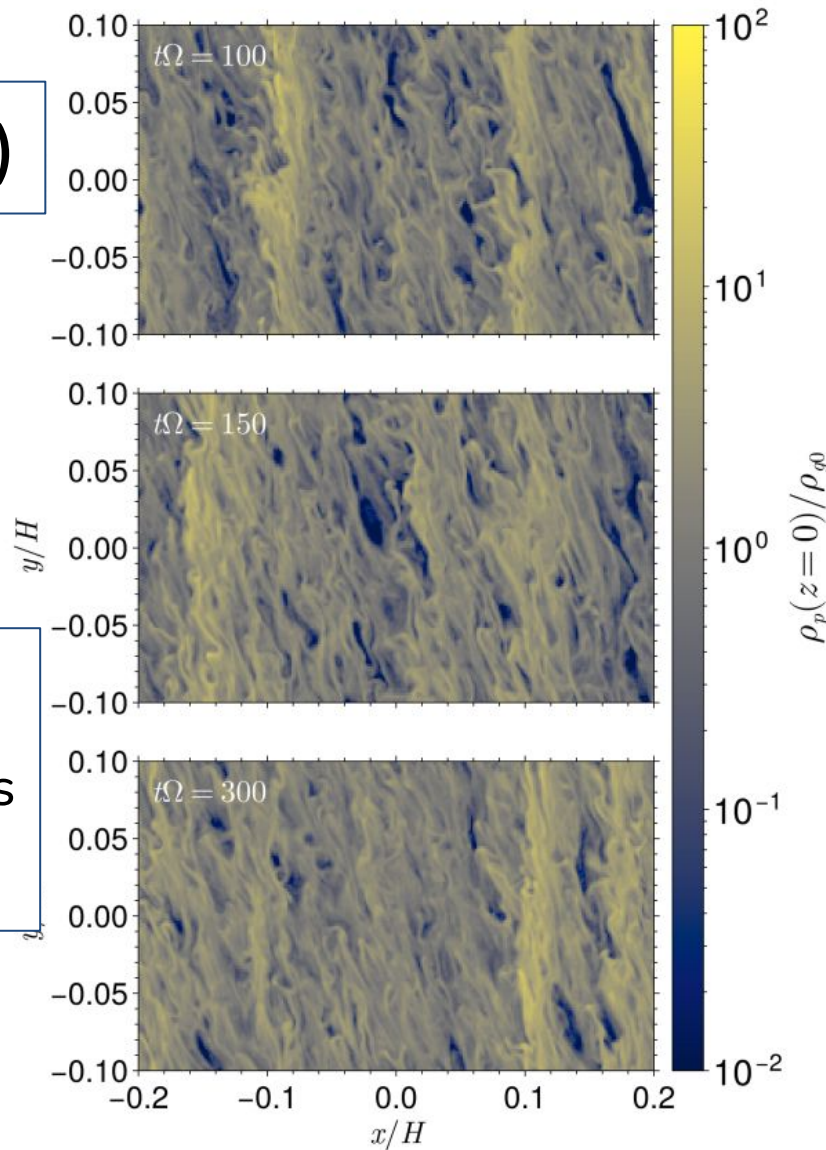
Step 2: SI concentrates dust

Carrera+ (2025a)

Used the 3D runs of Lim+ (2024) to fit dust/gas ratio inside SI filaments.

$\epsilon_{\text{SI}} \approx$ Chosen so that
~70% of dust in
the filament sees
 $\epsilon > \epsilon_{\text{SI}}$

$$\text{St} = 0.1, \quad Z = 0.02, \quad \alpha = 10^{-4}$$

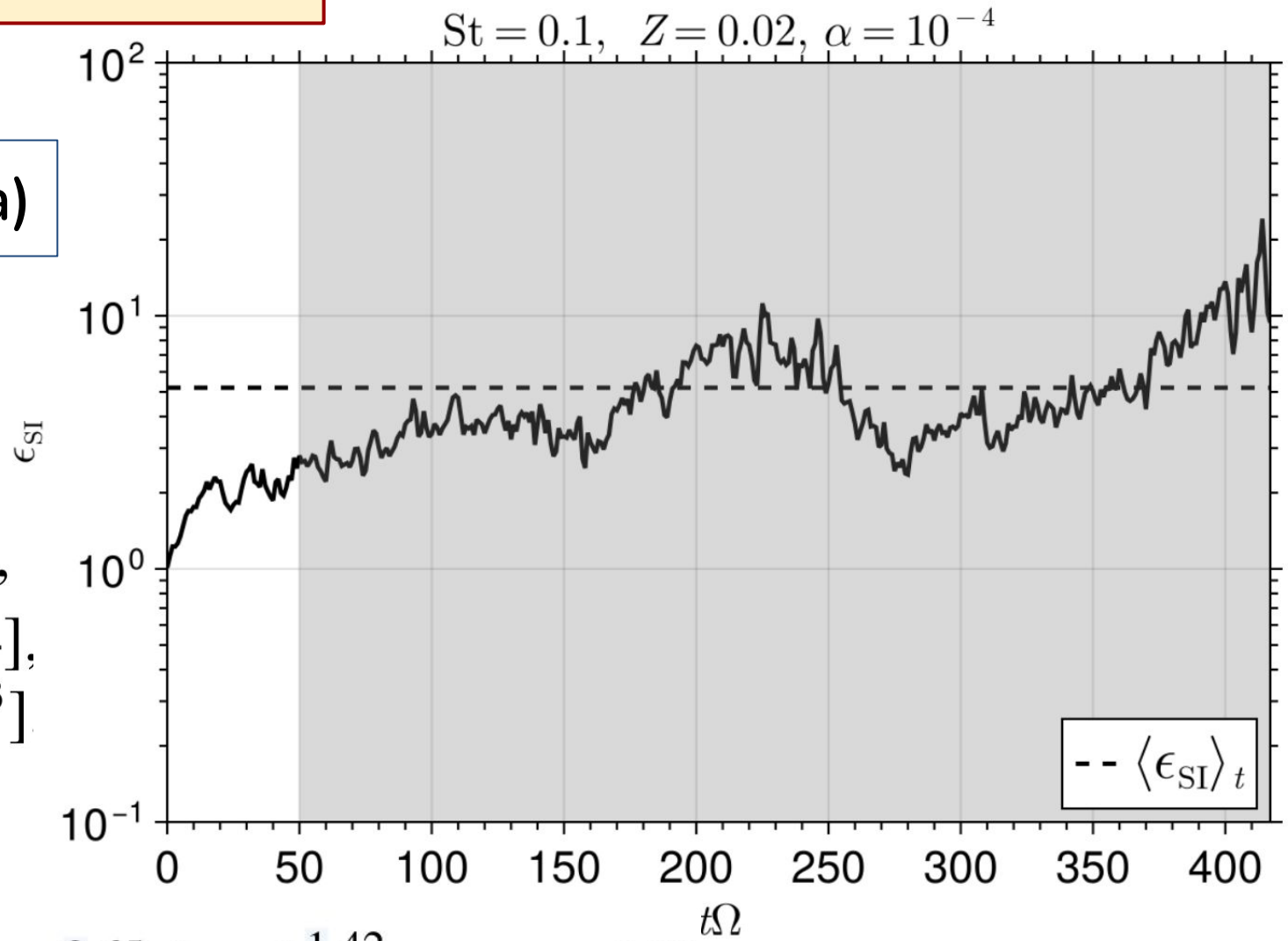


Solution 1: Positive Feedback: SI+Coagulation

Step 2: SI concentrates dust

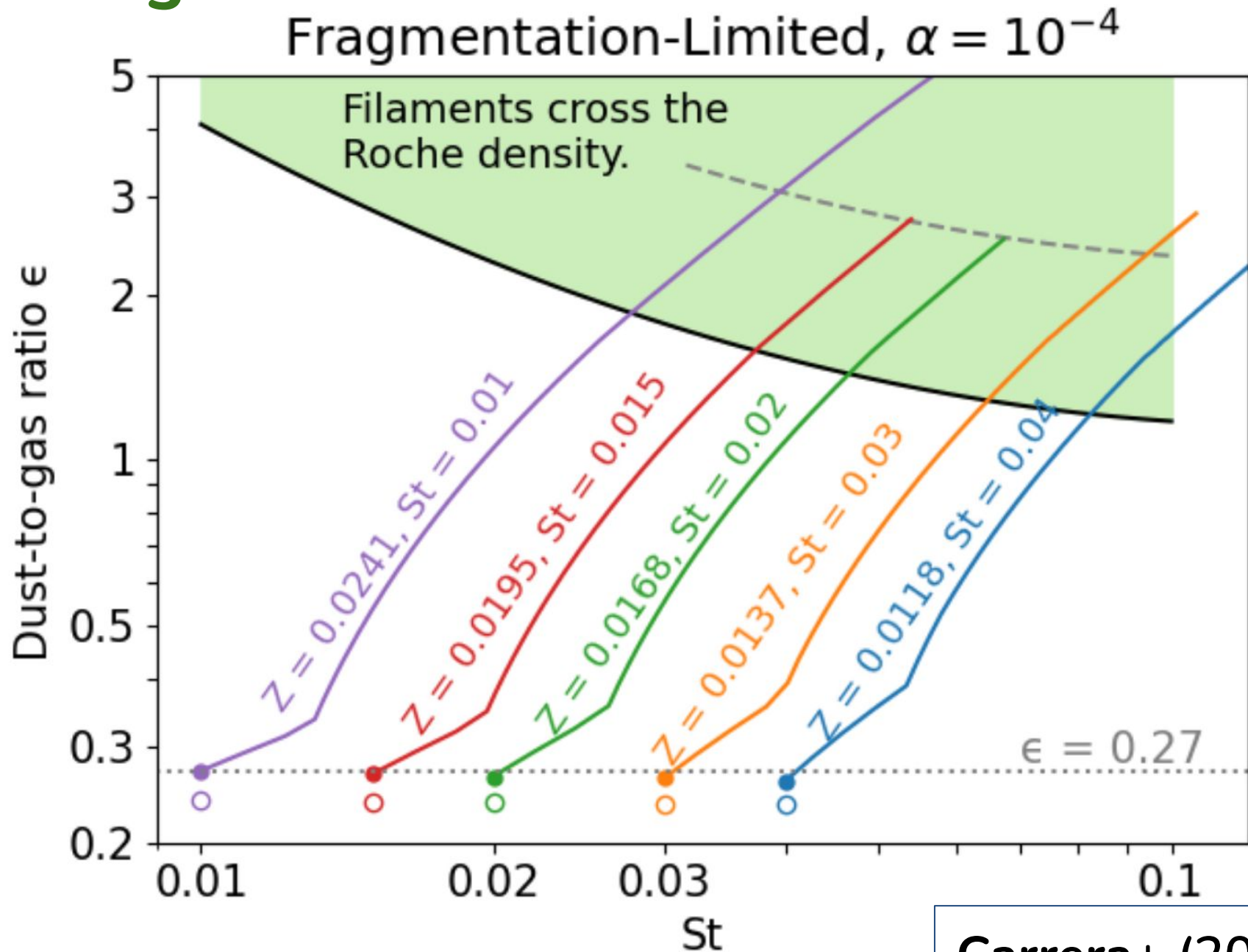
Carrera+ (2025a)

$$\begin{aligned} St &\in [0.01, 0.1], \\ Z &\in [0.015, 0.4], \\ \alpha &\in [10^{-4}, 10^{-3}] \end{aligned}$$



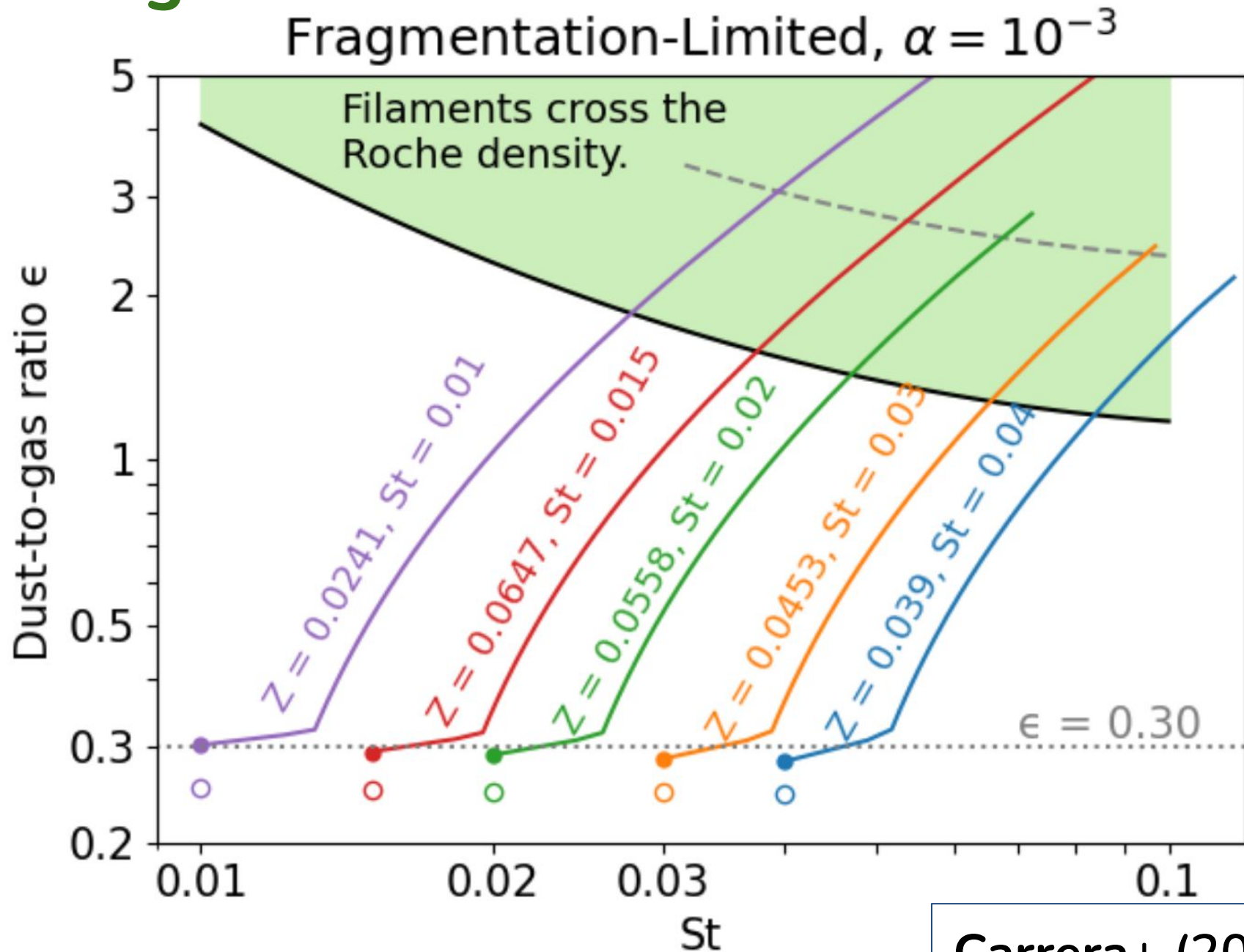
$$\epsilon_{SI} \approx 1.12 \left(\frac{Z}{0.01} \right)^{2.65} \left(\frac{St}{0.1} \right)^{1.42} \left(\frac{\alpha}{10^{-4}} \right)^{-1.38}$$

Solution 1: Positive Feedback: SI+Coagulation



Carrera+ (2025a)

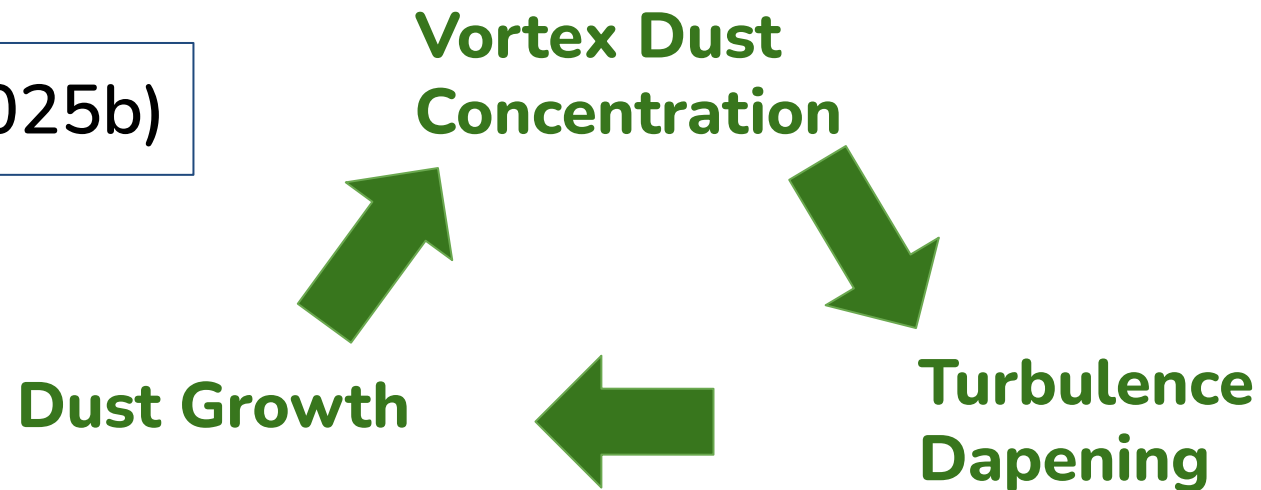
Solution 1: Positive Feedback: SI+Coagulation



Carrera+ (2025a)

Solution 2: Positive Feedback II: Vortex+Coag

Carrera+ (2025b)



Key insight:

Vortices also collect dust $\propto$ grain size. $\Rightarrow$ Similar feedback as with the streaming instability.

Model Ingredients:

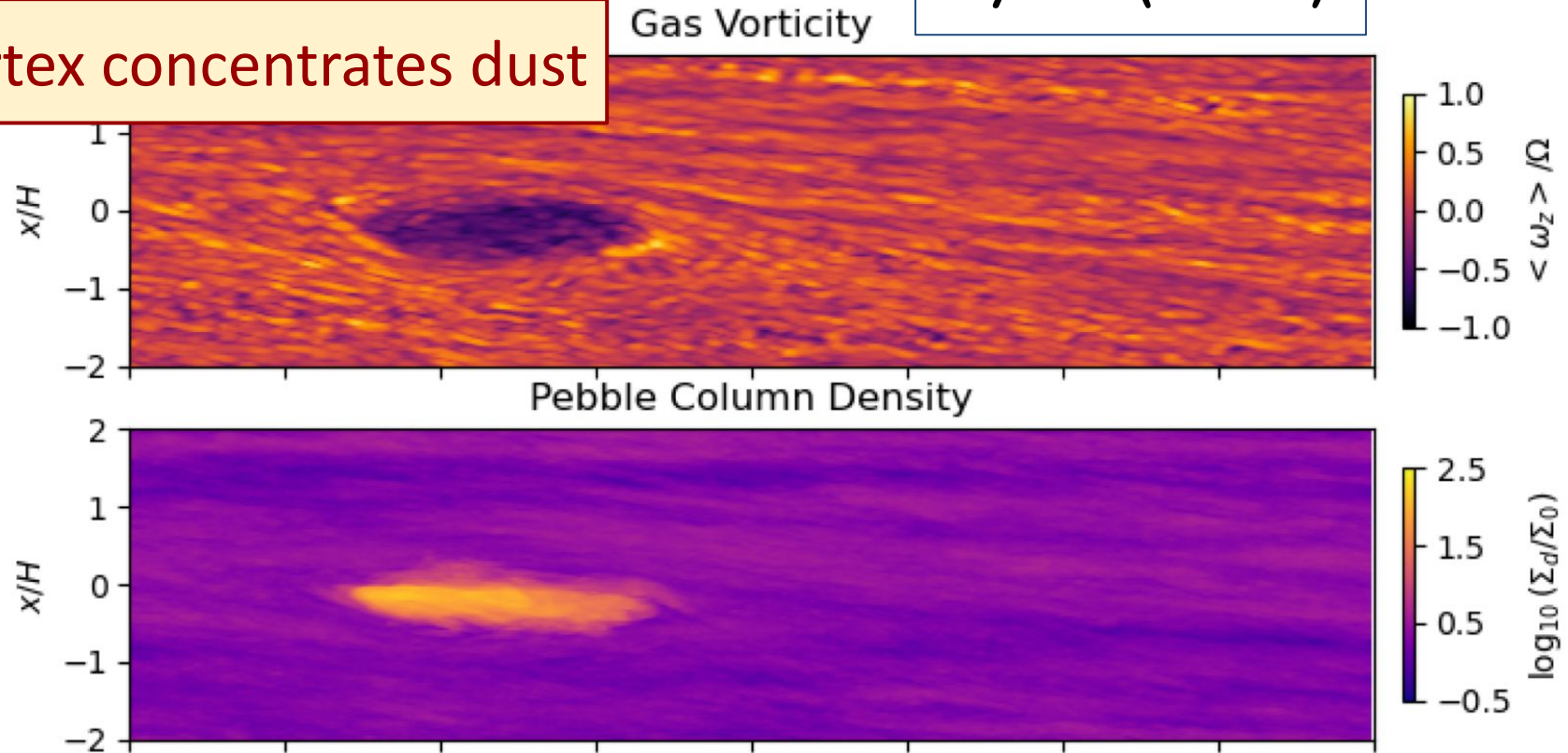
1. Vortex concentrates dust (analytic)
2. Feedback dampens turbulence (analytic)
3. Lower turbulence = less fragmentation (analytic)

Solution 2: Positive Feedback II: Vortex+Coag

Step 1: Dust dampens turbulence

Step 2: Vortex concentrates dust

Lyra+ (2024)



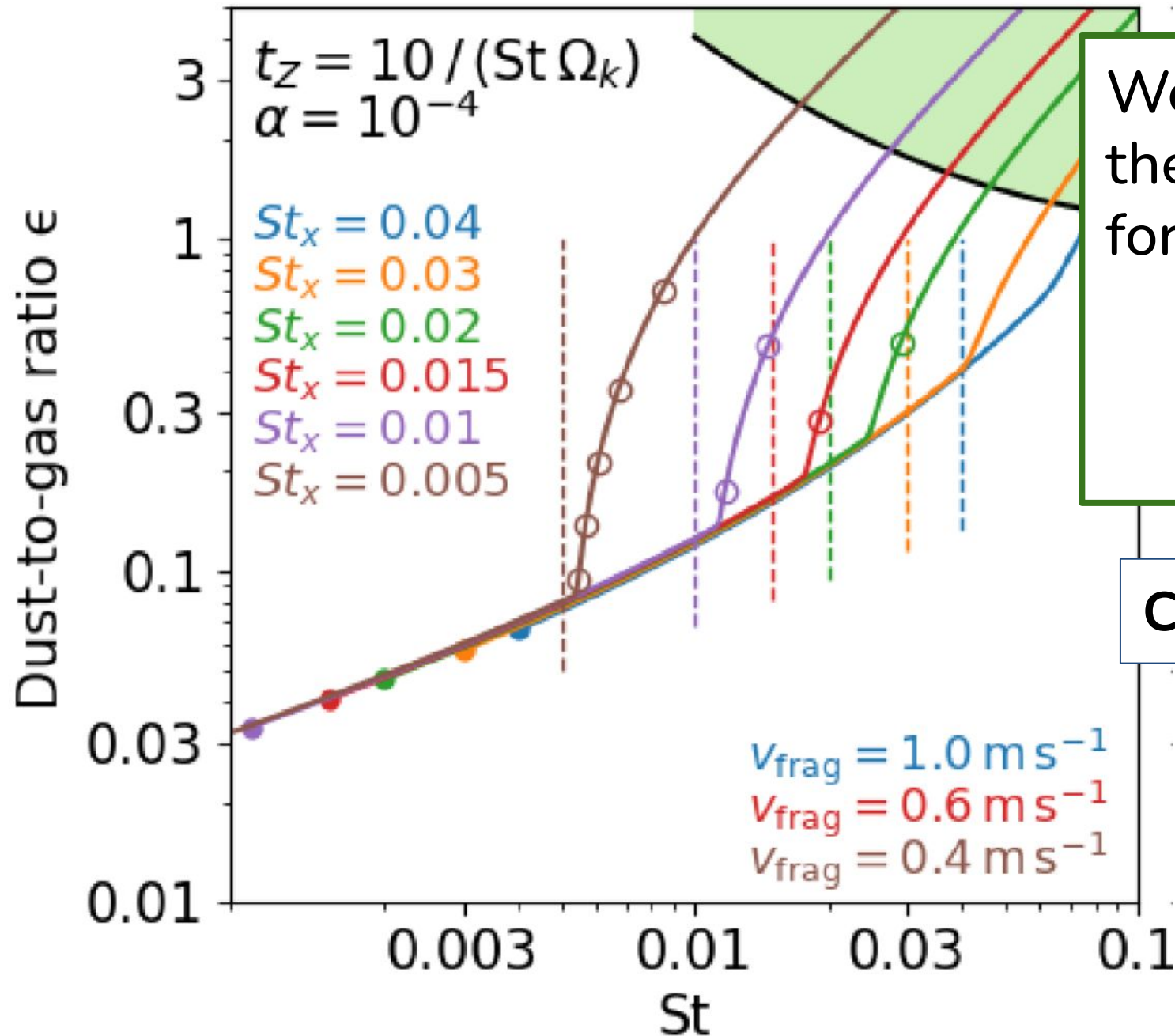
Vortex trapping: $Z_{\max} = Z_{\text{disk}} \left(1 + \frac{\text{St}}{\alpha} \right).$

Lyra+Lin (2013)

Trapping timescale: $t_Z \in \left[\frac{1}{\text{St} \Omega_K}, \frac{10}{\text{St} \Omega_K} \right].$

Carrera+ (2025b)

Solution 2: Positive Feedback II: Vortex+Coag



We ***always*** reach
the planetesimal
forming region for

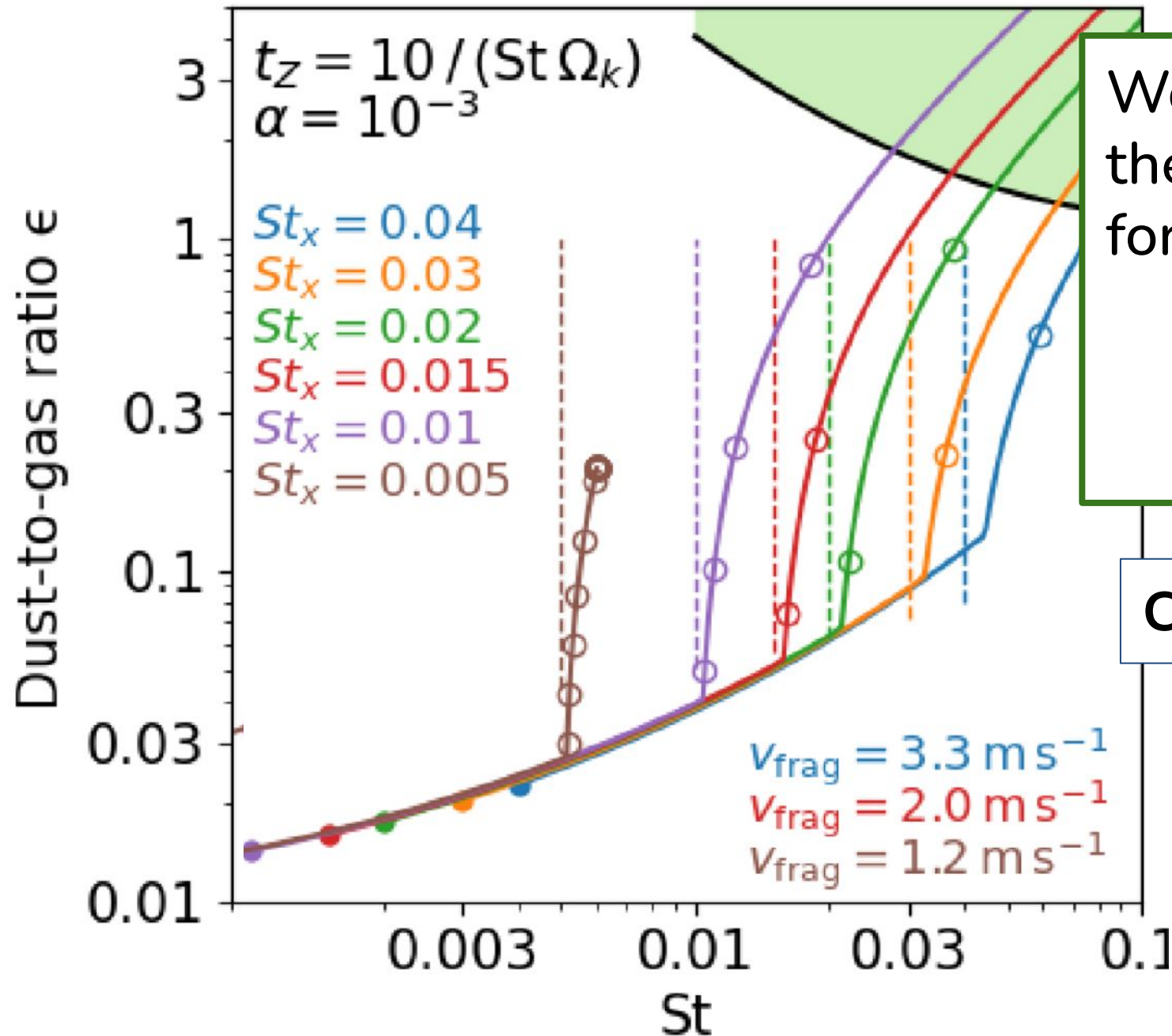
$$\mathbf{St} = \mathbf{0.01}$$

$$\mathbf{Z} = \mathbf{0.01}$$

$$\alpha \leq \mathbf{10^{-3}}$$

Carrera+ (2025b)

Solution 2: Positive Feedback II: Vortex+Coag



We **always** reach the planetesimal forming region for

- $St = 0.01$**
- $Z = 0.01$**
- $\alpha \leq 10^{-3}$**

Carrera+ (2025b)

Conclusions:

(1) We need to focus on small grains
($St \leq 0.01$)

- Dust growth models predict $St \leq 0.01$
- The SI struggles with $St \leq 0.01$

(2) We need to model the
Coagulation-Concentration Feedback Loop

- Particle feedback dampens turbulence
- Positive Feedback I: $SI \rightleftharpoons \text{Coagulation}$
- Positive Feedback II: $\text{Vortices} \rightleftharpoons \text{Coagulation}$

See also: Talks by Lin, Orkan, Debanjan, and Ryosuke